

LEZ 1

PRELIMINARIES (Recap)

PRELIMINARIES: $\rightarrow$ notations, Terminology

for all the course we approach LINEAR TIME INVARIANT system (LTI)

(later on see distributed control and communication issues)

In CONTINUOUS TIME framework:

state-space

$$\begin{cases} \dot{x} = Ax + Bu \\ y = Cx + Du \end{cases}$$

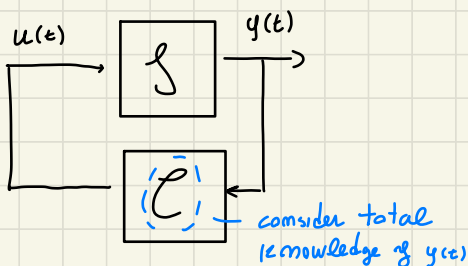
$$\begin{aligned} x &\in \mathbb{R}^n \\ u &\in \mathbb{R}^m \\ y &\in \mathbb{R}^p \end{aligned}$$

Dynamic model of the system

$\downarrow$
cont time
 $\downarrow$

BUT you can threat discrete time

FRAMEWORK, CONTROL:



$\hookrightarrow$ to compute state/output $\forall t$ - LAGRANGE EQUATION

given $x(t_0) = x_0$ $u(t), \forall t$:

$$x(t) = \underbrace{e^{A(t-t_0)} x_0}_{\text{FREE MOTION (due to } x_0)} + \underbrace{\int_{t_0}^t e^{A(t-\tau)} B u(\tau) d\tau}_{\text{FORCED MOTION (due to } u(t))}$$

formula used
 $\checkmark$ a lot in discrete time

plug in om $y(t)$
 $\downarrow$

$$y(t) = \underbrace{C e^{A(t-t_0)} x_0}_{\text{OUTPUT free motion}} + \underbrace{C \int_{t_0}^t e^{A(t-\tau)} B u(\tau) d\tau + D u(t)}_{\text{FORCED motion (due } u(t))}$$

EQUILIBRIA (points which remain constant in time once defined $u(t) = \bar{u}$)

IF fixed input constant $u(t) = \bar{u}$, if equilibrium

$$\dot{x} = 0 = A\bar{x} + B\bar{u} \rightarrow A\bar{x} = -B\bar{u} \quad \text{IF } A \text{ is invertible}$$

$\bar{x} = -A^{-1}B\bar{u}$ unique equilibrium!

(equilibrium can NOT exist, be ∞ ..)

matrix of complement algebraic

→ inverse of matrix

$$A^{-1} = \frac{1}{\det(A)} \text{Adj}(A)$$

of states = # eig values

A is invertible $\Leftrightarrow \det(A) \neq 0$

$$\hookrightarrow \det(A) = \prod_{i=1}^n \lambda_i \neq 0$$

$\Updownarrow$

NO eigen value = 0

CNS for existence of equilibrium $\forall u(t)$

PROPERTY of EQUILIBRIA..

(stable, asympt. stable, unstable etc..)

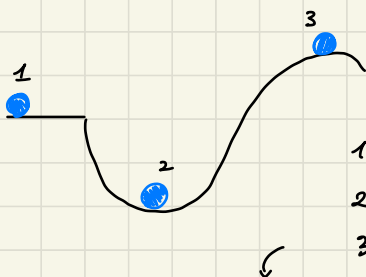
STABILITY: ("of each motion") $\rightarrow$ in LTI its a property of the considered system

(property of matrix A)

Example

① := 3 equilibrium points of system.

$\hookrightarrow$ BUT different, if perturbed, changing ball position, for small perturbation



1) remains in neighbourhood **stable** (NOT asympt)

2) perturbed move around and stop again in initial (asympt stable)

3) go far away from initial condition (**unstable**)

different equilibria on different points is impossible for LTI, but in NON LINEAR syst.

$\downarrow$

to check for stability we look eigenvalues:

Theorem: $\mathcal{S}$ is asymptotically stable IF and only IF all eig(A) has $\text{Re} < 0$

IFF $\text{Re}(\lambda_i) < 0 \quad \forall i \rightarrow \parallel A \text{ is Hurwitz} \parallel$ (continuous time syst)
stable

LINEAR system... equivalent stability definition independently from λ_i definition...



from inequality: LINEAR MATRIX inequality

DEFINITION: $P = P^T$, $P \in \mathbb{R}^{n \times n}$ (symmm matrix given)

Example
symmm
MATRIX
 $\begin{bmatrix} 1 & -2 \\ -2 & 5 \end{bmatrix}$

• P is positive definite ($P > 0$) $\Leftrightarrow x^T P x > 0 \quad \forall x \neq 0$

example:

1) $\begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix} = P$ yes, positive definite $\rightarrow x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \rightarrow x^T P x = \begin{bmatrix} x_1 & x_2 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = x_1^2 + 2x_2^2 > 0 \quad \forall x \neq 0$
↓
to be positive definite elements can be also negative

2) $\begin{bmatrix} 1 & -1/4 \\ -1/4 & 1 \end{bmatrix} \quad x^T P x = \begin{bmatrix} x_1 & x_2 \end{bmatrix} \begin{bmatrix} 1 & -1/4 \\ -1/4 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = x_1^2 - \frac{1}{2} x_1 x_2 + x_2^2$

it can be neg for some x_1, x_2 ?

IF collect $1/4$

↓
to be def positive is NOT necessary to have all elements positive

$$\begin{aligned} & \frac{1}{4} (4x_1^2 - 2x_1x_2 + 4x_2^2) = \\ & \frac{1}{4} (x_1^2 - 2x_1x_2 + x_2^2 + 3x_1^2 + 3x_2^2) \\ & \frac{1}{4} \left[\underbrace{(x_1 - x_2)^2}_{\geq 0 \text{ always}} + \underbrace{3x_1^2 + 3x_2^2}_{\geq 0 \text{ always}} \right] > 0 \quad \forall x \neq 0! \end{aligned}$$

so it is still pos def

• P is (positive semi definite) $P \geq 0 \Leftrightarrow x^T P x \geq 0 \quad \forall x \neq 0$

Example

$P = \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} \geq 0 \rightarrow \text{fact} : x^T P x = \begin{bmatrix} x_1 & x_2 \end{bmatrix} \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = x_1^2 - 2x_1x_2 + x_2^2 = (x_1 - x_2)^2 \geq 0$
↓ similarly
if $x_1 = x_2$ its 0! it's semi def pos

• P is (Negative definite) $P < 0 \Leftrightarrow -P > 0$ ($-P$ pos. def)

• P is (Negative semi definite) $P \leq 0 \Leftrightarrow -P \geq 0$ ($-P$ semi pos. def)

PROPERTY: IF $P > Q$ ($P = P^T$, symm) $\Rightarrow$ All eigenvalues are real and positive

taking $0 < \lambda_{\min} < \dots < \lambda_{\max}$ (ordering eig values)

$\hookrightarrow$ and $\forall x \neq 0 \quad \lambda_{\min} \|x\|_2^2 \leq x^T P x \leq \lambda_{\max} \|x\|_2^2$

Theorem: (it can be proven:)

IF $\exists P = P^T$ (symm), $P > 0$: $A^T P + P A < 0 \iff A$ is Hurwitz stable

(look stability from algebraic property of the system

without check eigenvalues) $\rightarrow$ related still to eig values

(useful technique to develop strong control both for LTI, LTV syst)

$\hookrightarrow$ stability result for wider class of system than only on LTI!

(related to eig) CNS!

$\downarrow$ I can provide, $\nexists P$ its unstable

idea behind proof:

$(\Leftarrow)$ IF A is H. stable $\Rightarrow \exists P = P^T$ s.t. $P > 0$
 $A^T P + P A < 0$

start defining arbitrary matrix $Q = Q^T > 0$

and,

IF we take $P = \int_0^{+\infty} e^{A^T t} Q e^{A t} dt > 0$ $\nearrow$ this satisfy this equation! and ∞

we can

prove integral

well defined,

due to H. stability $e^{A^T t}, e^{A t}$ are exponentially convergent

putting dom the inequality:

$\hookrightarrow$ that's because eig one with neg (Re part!) $\rightarrow$ implicitly use eig

$$A^T \int_0^{+\infty} e^{A^T t} Q e^{A t} dt + \int_0^{+\infty} e^{A^T t} Q e^{A t} dt A$$

show that it is $< 0 \dots$

$$= \int_0^{+\infty} \underbrace{(A^T e^{A^T t} Q e^{A t} + e^{A^T t} Q e^{A t} A)}_{\frac{d}{dt} (e^{A^T t} Q e^{A t})} dt = \int_0^{+\infty} \left(\frac{d}{dt} e^{A^T t} Q e^{A t} \right) dt =$$

$$= e^{A^T t} Q e^{A t} \Big|_0^{+\infty} = \underbrace{0}_{\substack{\text{@ } t=+\infty \\ \text{both } = 0 \\ \text{because} \\ \text{decreasing} \\ \text{exponentially}}} - Q \rightarrow A^T P + P A = -Q < Q$$

$Q > 0 \Leftrightarrow -Q < Q$

SO FROM H-stable matrix
you find a P satisfy that
condition always!

($\Rightarrow$) IF $\exists P = P^T > Q$

such that $(A^T P + P A < Q) \Rightarrow A$ is H. stable

easy to show... $\hookrightarrow$ from here, take an eig value of A and
its eig vector:

$\lambda, v :=$
eig value eig vector

$$A v = \lambda v \text{ (by definition)}$$

multiply by $v^T v$

$$v^T (A^T P + P A) v = \underbrace{v^T A^T P v}_{(A v)^T = (\lambda v)^T} + \underbrace{v^T P A v}_{\lambda v} \dots$$

notice: on complex
vector $v \in \mathbb{C}^n \rightarrow$ hermit transposition

$$\left(\begin{array}{l} v^T = \text{TRANSPOSE} + \text{complex conj} \\ \uparrow \\ \text{transposed and any values is} \\ \text{transposed and complex conjugate} \end{array} \right)$$

example

IF taking

$$P = 2, P > Q \text{ pos. def}$$

taking $x = i \rightarrow x^T P x = (i) P i = -P = -2 < Q$?! don't correspond to pos. def definiton

on complex conj transposition:

$$\Rightarrow \text{hermit transposition } x^T P x = (i^*) P i = -i P i = P = 2 > Q \text{ OK!}$$

, we don't transpose this way!

→ by hermit transpose

complex conj of λ

$$\lambda^* v^T = (\lambda v)^T$$

$$\dots = \lambda^* v^T P v + v^T P v \lambda = v^T P v (\lambda + \lambda^*) =$$

for example

↑ valid imaginary!

$$\lambda = 2 + j : \lambda + \lambda^* = 2 + j + 2 - j = 4 = 2 \operatorname{Re}(\lambda)$$

$$= \underbrace{2 \operatorname{Re}(\lambda)}_{\substack{\text{this} \\ \text{must be} \\ 2 \operatorname{Re}(\lambda) < 0 \rightarrow \operatorname{Re}(\lambda) < 0}} \underbrace{v^T P v}_{\substack{> 0 \text{ because} \\ P > 0}} < 0 \text{ from hypothesis!}$$

A is an

valid $\forall \lambda \Rightarrow$ H-stable matrix!

□

Remark...

$A^T P + P A < 0$ is the LYAPUNOV inequality (matrix inequality)

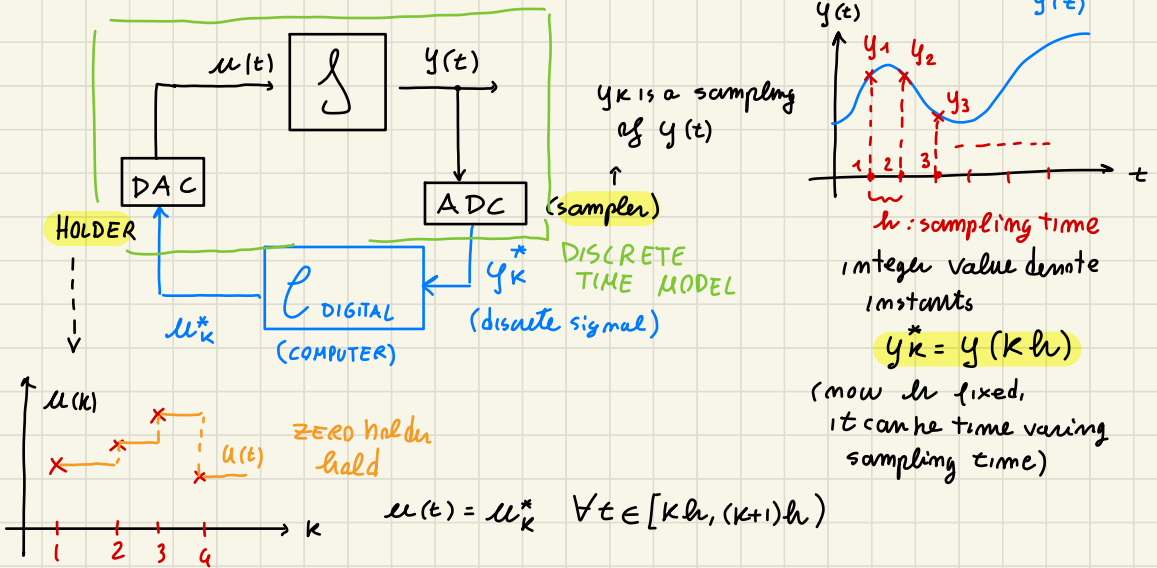
IF we compute $V(x) = x^T P x$ $\dot{x} = Ax$ (Autonomous syst)

from Lyapunov Theorem $\dot{V}(x) = \dot{x}^T P x + x^T P \dot{x} = x^T A^T P x + x^T P A x =$

$$= x^T \underbrace{(A^T P + P A)}_{\substack{\text{should be neg def} \\ \text{from hyp}}} x < 0 \text{ because its } \dot{V} < 0$$

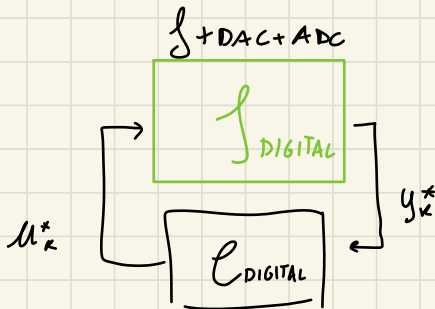
↓
coincide with Lyapunov theory

DISCRETE TIME MODELS... Looking from digital control P.O.V



to design $\mathcal{C}_{\text{DIGITAL}}$, we have 2 ways:

- 1) consider $\mathcal{G}$, design $\mathcal{C}$ and discretize it! (classical way)
- 2) consider overall system as box with $u_k^*(z)$ and $y_k^*(z)$ from discrete time model $\rightarrow$ and design $\mathcal{C}_{\text{DIGITAL}}$ for this $\mathcal{G}_{\text{DIGITAL}}$



to design $\mathcal{C}_{\text{DIGITAL}}$.

1) • Design $\mathcal{C}_{\text{CONT TIME}} \rightarrow$ DISCRETIZE THE CONTROLLER

2) • use the discrete time model

$\downarrow$
Design $\mathcal{C}_{\text{DIGITAL}}$

more flexible way, you can use MPC, ecc..

But you have to find the model of system discrete?



How to derive the discrete time model from the continuous time one?

(I use Lagrange equation!)



from random instant $k \Rightarrow t = kh$ model which describe how x_{k+1} depends on x_k ?

$$x_{k+1}^* = x((k+1)h) = \dots$$



using Lagrange equation

from $t_0 = kh$ and $t_1 = (k+1)h$

$$\dots = e^{Ah} \underbrace{x(kh)}_{x_k^*} + \int_{kh}^{(k+1)h} e^{A((k+1)h-\tau)} B u(\tau) d\tau$$

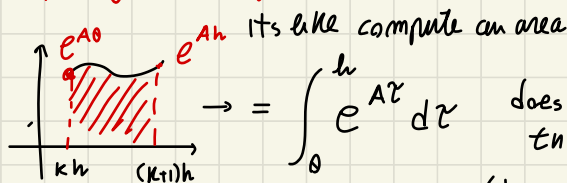
↓ goto..
 $k+1$

from the HOLDER, $\forall$ time window... $u_k^* = u(\tau)$ for $\tau \in [kh, (k+1)h)$

$$= e^{Ah} x_k^* + \int_{kh}^{(k+1)h} e^{A((k+1)h-\tau)} d\tau B u_k^*$$

computing an integral

that's already a discrete time description of system



its like compute an area

does NOT depend on k
this integral!
(time invariant)

$$x_{k+1}^* = \underbrace{\left[e^{Ah} \right]}_F x_k^* + \underbrace{\left[\left[\int_0^h e^{A\tau} d\tau \right] B \right]}_G u_k^* = F x_k^* + G u_k^*$$

common discrete time model

how to compute F, G ↑

exact model $\Leftarrow$ model!

EXACT model, NOT approx

using mathematic, NO MISMATCH! but don't describe what happens between the samples...

$$x_{k+1}^* = F x_k^* + G u_k^*$$

$$\begin{cases} F = e^{Ah} \\ G = \left(\int_0^h e^{A\tau} d\tau \right) B \\ \Gamma(h) \end{cases}$$

21/09

once found discrete time realization:

$$x_{k+1} = F x_k + G u_k$$

as discretization of the
cont. time model

(there are correspondence to the
cont-time one)

with $F = e^{Ah}$

$$G = \left[\int_0^h e^{A\tau} d\tau \right] B$$

$\Gamma(h)$

↓

PROPERTY:

solution knowing

initial state $x(t_0) = x_0$

u_k (values taken in discrete time) → injected by DAC

Recursively $x_k = x(kh) = F^k x_0 + \underbrace{\sum_{i=1}^k F^{k-i} G u_{k-i}}_{\substack{\text{FREE} \\ \text{MOTION}}}$

since linear system..

divide the system
motion into free/forced

↓ comparing respect
continuous time

Free motions

• cont. time: $x(t) = e^{At} x_0 \rightarrow x(kh) = x_k = (e^{Ah})^k x_0$

• discrete time: $x_k = F^k x_0$ ← same formula!

(Sample & Hold is the
only assumption of
discretized system)

↑
different formulation

of same model in different framework usage

Discrete: in network control (digital control)

NOT an
approx
model,
perfect syst.
dynamic

discrete time model
is NOT an approx, but
exact model!

Equilibria keep constant input $u_k = \bar{u}$ (yet valid in assumption)

$$x_{k+1} = x_k = \bar{x} \Rightarrow \bar{x} = F \bar{x} + G \bar{u}$$

$$(I - F) \bar{x} = \underbrace{G \bar{u}}_{\text{known}} \quad \text{linear equation}$$

(unique solution $\Leftrightarrow (I - F)$ is invertible)

IFF $(I - F)$ is invertible :

$$\bar{x} = (I - F)^{-1} G \bar{u}$$

In continuous time:
(condition)

A invertible $\Leftrightarrow \exists!$ equilibrium

In discrete time

$I - F$ invertible

relationship?
 $\downarrow$
YES!

We can show this relation in our definition

in diag all element = 1

$$I - F = I - e^{A h} \quad \text{invertibility when}$$

$\updownarrow$
eig of A are $\neq 0$

$\updownarrow$
 A is invertible!

$$\lambda(\underbrace{I - e^{A h}}_{\neq 0}) \neq 0$$

$$1 - \lambda(e^{A h}) = 1 - e^{\lambda(A) h} \neq 0$$

$$\Leftrightarrow e^{\lambda(A) h} \neq 1$$

$$\ln(e^{\lambda(A) h}) \neq \ln 1$$

$$\lambda(A) \neq 0 \Leftrightarrow \lambda(A) \neq 0$$

$\neq 0$ always

(the two models are
consistent!
equivalent formulation)

Stability $\rightarrow$ in discrete time model, theory related to eig values of F ($\lambda(F)$)
checking absolute value

Theorem

The discrete-time system is asymptotically stable
(all system motion have asympt. behav.)

$$|\lambda_i(F)| < 1 \quad \forall i=1, \dots, n$$

$\downarrow$
consistent with
continuous time requirements

$\Rightarrow F$ is Schur stable

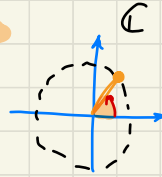
$$\lambda_i(F) = \lambda_i(e^{Ah}) = e^{\lambda_i(A)h} = \underbrace{e^{\operatorname{Re}(\lambda_i(A))h}}_{\substack{\text{Re part} \\ \text{AMPLITUDE}}} \cdot \underbrace{e^{j \operatorname{Im}(\lambda_i(A))h}}_{\substack{\text{it represent} \\ \text{a point on} \\ \text{unit circle} \\ \text{complex plane } e^{j\theta}}}$$

to compute $|\lambda_i(F)| =$

$$= |e^{\operatorname{Re}(\lambda_i(A))h}| \cdot |e^{j \operatorname{Im}(\lambda_i(A))h}| = 1$$

$$= |e^{\operatorname{Re}(\lambda_i(A))h}| < 1 \iff \operatorname{Re}(\lambda_i(A)) < 0$$

IF negative
exponent
($h > 0$)



$| \cdot | = 1$ (point on
circumference)

$\downarrow$
like done in cont. time, alternative checking asympt stability ...
(without checking eig values:)

Theorem: (similar to cont. time)

$$F \text{ is Schur stable} \iff \exists P = P^T > 0 : F^T P F - P < 0$$

(similar to cont.
time one, but different
inequality because discrete time)

$\downarrow$
I can highlight
this relation with
eig of F matrix.. underline relationship $\rightarrow$ sketch
proof of theorem

PROOF

(knowing F shun stable)

($\Rightarrow$) from corresponding cont. time one: $P = \int_0^{+\infty} e^{A^T \tau} Q e^{A \tau} d\tau$

corresponding in discrete time:

Where $Q > 0$ ARBITRARY, $Q = Q^T$

$e^{A\tau}$: sampling on many $\tau = kh$ (sampling time)

$$\hookrightarrow e^{AKh} = (e^{Ah})^k = F^k$$

and $\int_0^{+\infty} \dots \rightarrow \sum_{k=0}^{+\infty}$
infimite sum

using this redefined P

$$P = \sum_{k=0}^{+\infty} (F^T)^k Q F^k$$

$(Q + F^T Q F + (F^2)^T Q F^2 \dots)^T$
 $Q^T + F^T Q F + \dots$ symmetric
 $P = P^T$ Pas sum of symmetric!

P is defined well since F is shun stable

now let's check what value assume

$$F^T P F = F^T \left(\sum_{k=0}^{+\infty} (F^T)^k Q F^k \right) F = \dots$$

$$\dots = \sum_{k=0}^{+\infty} (F^T)^{k+1} Q F^{k+1} = \underbrace{Q + F^T Q F + (F^T)^2 Q F^2 + (F^T)^3 Q F^3 + \dots}_{P} - Q$$

$$= P - Q$$

(summ/remove for my result!)

$$\downarrow$$
$$F^T P F = P - Q \Rightarrow F^T P F - P = -Q < 0 \text{ (neg def - Q)}$$

OK inequality

holds if F shun stable

(guarantee P were defined)

($\Leftarrow$) existence of that P + inequality leads to F is stable... (as done in Cont. time)

$$(P > 0), \quad F^T P F - P < 0$$

$$v^T P v > 0 \quad \forall v \text{ vector}$$

$$(\lambda v)^T F^T P F (\lambda v) - v^T P v < 0$$

$$(\lambda v)^T P (\lambda v) - v^T P v < 0$$

$$\lambda^* \lambda v^T P v - v^T P v < 0$$

$\lambda^* \lambda = |\lambda|^2$ in fact, for example:

$$\lambda = 2 + j \quad \lambda^* = 2 - j : \quad \lambda \lambda^* = (2 + j)(2 - j) = 2^2 + 1^2 = 5 = |\lambda|^2$$

$$(|\lambda|^2 - 1) \underbrace{v^T P v}_{> 0 \text{ hypothesis}} \stackrel{\text{by hypothesis}}{<} 0$$

must be negative!

$$\Rightarrow |\lambda|^2 - 1 < 0 : \quad |\lambda|^2 < 1$$

which is shown
stability of matrix F !

↓

by that Theorem, we don't have to check for eig, but search for a matrix P ...

↳ advantages $\Rightarrow$ lightweight numerical method,
usefull also for controller design!

↓

and this method by inequality can be used for
linear time VARIING system! (differently from eig values
analysis valid only for time-invariant)

$$v^T (F^T P F - P) v < 0 \quad \forall v$$

taking vector v as eig vector of F
corresponding to eigenvalue λ

$$F v = \lambda v \text{ because definition}$$

$$(\lambda v)^T = \lambda^* v^T$$

↑
transposing a
complex vector...
we have to transpose
and use complex conjugate!

(to assess discrete time stability, 2 ways numerically:)
to check asymptotic stability:

1) compute eigenvalues of F usefull method
in MATLAB

2) check for $P > 0$ avoid explicit eig values..
 $F^T P F - P < 0$

3) From the characteristic polynomial?

(in continuous time: other criterion of stability check, ROUTH method ecc...)
↳ some similar in discrete time?

↓

JURY

CRITERION

(simplify stability check from characteristic polynomial)
analyze parameter of it..

↓

(not used in this course!) usually used for II ORD system..

Steps of CRITERION

1) table definition: JURY TABLE

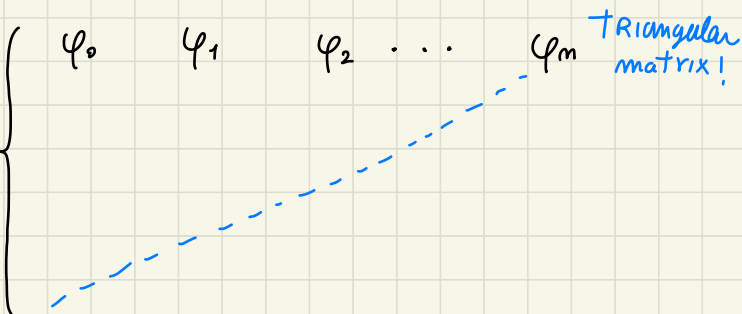
2) apply rule, checking sign of first column table...

① **JURY TABLE**

from characteristic polynomial $\det(\lambda I - F)$ polynomial of order n

$$P_F(\lambda) = \varphi_0 \lambda^n + \varphi_1 \lambda^{n-1} + \dots + \varphi_n$$

TABLE: from first line..

	φ_0	φ_1	φ_2	$\dots$	φ_n
$(n+1)$ rows					

Iterative computation of Rows, 3RD based on 2ND
 G^{TH} // // 3RD
 iteratively

$$\begin{array}{cccccc} K^{TH} - \text{ROW} & K_1 & K_2 & \dots & K_{v-1} & K_v \\ (K+1)^{TH} - \text{ROW} & \ell_1 & \ell_2 & \dots & \ell_{v-1} & 0 \end{array}$$

general element

$$\ell_i = \frac{1}{K_1} \det \begin{bmatrix} K_1 & K_{v-i+1} \\ K_v & K_i \end{bmatrix}$$

IF $K_1 = 0$,
 you cannot compute!

IF $K_1 = 0$: the table is
 NOT well defined

after defining
 1st Row from $\ell_0, \ell_1, \dots$
 the others are iterative

② apply the JURY RULE to check stability

F is SHUR stable $\Leftrightarrow$ $\begin{cases} \cdot \text{JURY TABLE is well defined} \\ \quad (\text{defined until last Row}) \\ \cdot \text{all elements of the FIRST column} \\ \quad \text{of the JURY table are concordant} \\ \quad (\text{same sign}) \end{cases}$

(used in
 network

control system exercise,

2D dynamic $\rightarrow$ inadequate time used for stability)

$\downarrow$

2nd ORDER problem:
 System

$\downarrow$

Remark: JURY CRITERION for $m=2$

$F \in \mathbb{R}^{2 \times 2} \rightarrow$ char. polynomial of order 2

$$P_F(\lambda) = \lambda^2 + a\lambda + b \quad (\varphi_0 = 1 \text{ always in this cases..})$$

JURY TABLE: (3 ROWS by 3 COL.)

$$\begin{array}{ccc} 1 & a & b \\ x_1 & x_2 & 0 \\ y_1 & 0 & 0 \end{array}$$

↑
triangular,
y₁ last element
to compute!

↓ Here $V=2$

$$y_1 = \frac{1}{x_1} \det \begin{bmatrix} x_1 & x_2 \\ x_2 & x_1 \end{bmatrix} = \frac{1}{x_1} (x_1^2 - x_2^2) =$$

↑
K₁ on that row

$$= \frac{1}{(1-b^2)} \left(\underbrace{(1-b^2)^2}_{(1-b)(1+b)} - a^2(1-b)^2 \right) = \frac{(1-b)^2}{1-b^2} \left[(1+b)^2 - a^2 \right]$$

↓

NOW apply JURY RULE to my TABLE! check 1st column 1, x₁, y₁
CONDITION (CNS) is that:

1, x₁, y₁ are concordant because $1 > 0$: $x_1 > 0$
 $y_1 > 0$

$$x_1 = 1 - b^2 > 0 \Leftrightarrow b^2 < 1 \Leftrightarrow |b| < 1 \rightarrow -1 < b < 1$$

$$y_1 > 0 \text{ (hard to check..)} = \frac{(1-b)^2}{1-b^2} \left((1+b)^2 - a^2 \right) \text{ Just check here for positiveness}$$

positive for sure!

$$(1+b)^2 - a^2 > 0$$

↑

← $-1 < b < 1$, $(1-b) \neq 0$ always, $(1-b)^2 > 0$
 $(1-b^2) > 0$ because as before!

$$-(1+b) < a < (1+b)$$

conditions to check for stable matrix
in 2nd order system from char polynomial

Reminder: Lyapunov inequality: $F^T P F - P < 0$ provided $P = P^T > 0$

is it related to the Lyapunov Theorem for discrete time model?

YES! ↓

in fact: in discrete time model cns of Lyap for stability of a system

$x_{k+1} = F x_k$ is asymp. (exponential) stable

IFF $\exists V(x_k) > 0 : V(x_{k+1}) - V(x_k) < 0 \quad \forall k \neq 0$

↓

we can define Lyap function as such that $V(x_k) = x_k^T P x_k > 0$ (because $P > 0$)

$$\text{and } V(x_{k+1}) = (F x_k)^T P (F x_k) = x_k^T F^T P F x_k$$

↓

$$V(x_{k+1}) - V(x_k) = x_k^T F^T P F x_k - x_k^T P x_k = x_k^T \underbrace{(-P + F^T P F)}_{< 0} x_k < 0 \quad \text{OK!}$$

(that's why called "Lyap" inequality)

↓

some usefull expressions:

$$P > 0 \text{ (positive definite)}$$

$$A^T P + P A < 0$$

$$F^T P F - P < 0$$

↓

this MATRIX inequalities fall into categories of

LINEAR MATRIX INEQUALITIES
(LMI)

(not used in control)

↓

equivalent to stability conditions,
so NOT used for control design
imposing stability/robustness/
optimality ecc..

how is constituted a MI to be called LINEAR
↓

the unknown matrix

(P) for which you have to check for,

all expressions perfectly linear in P!

OR "AFFINE" is enough $P \succ Q$ Affine inequality
→ still called LMI

⇔ LMI

AFFINE INEQUALITIES in P

↳ lot used in LMI feasibility problems :

(just have inequality without cost functions, just check if P exist)
like in STABILITY check
JUST SEE IF P EXIST! → JUST if
there is a solution fulfilling that
(analysis case)

$$f_1(P) > 0$$

⋮

$$f_m(P) > 0$$

↓
widely used in
stability analysis
(no optimize)

OR in LMI Optimization Problems

↓
Holds when doing controllers
design, for example in LQ control
try to optimize using some performance
(design problems)

$$\begin{aligned} & \min \underbrace{a \text{ linear function of } P}_{J(P)} \\ & \min J(P) \leadsto \text{cost linear in } P \\ & \text{s.t. } f_1(P) > 0 \\ & \quad \vdots \\ & \quad f_m(P) > 0 \end{aligned}$$

↳ learn how to write CI feasibility/optimization
in MATLAB,
and using YAMP software



Write optimization problem in SW,
how to cast this optimization problem!